

Integration WS 1

Evaluate Q1-10

1. $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cot x \, dx$

2. $\int_1^e \frac{1}{x} \cos(\ln x) \, dx$

3. $\int_0^2 \frac{1}{4+x^2} \, dx$

4. $\int_{\sqrt{2}}^3 \frac{1}{\sqrt{x^2-1}} \, dx$

5. $\int_0^2 \frac{1}{\sqrt{x^2+4}} \, dx$

6. $\int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-4x^2}} \, dx$

7. $\int_0^{\frac{\pi}{6}} \tan 2x \sec 2x \, dx$

8. $\int_0^{\ln 3} \frac{e^x}{1+e^x} \, dx$

9. $\int \frac{x-1}{x^2+1} \, dx$

10. $\int \frac{x(2x+1)}{x+1} \, dx$

11. a. By using the substitution $x = -u$, show

$$\text{that } \int_{-a}^0 f(x) \, dx = \int_0^a f(-x) \, dx$$

b. Deduce that $\int_{-a}^a f(x) \, dx = \int_0^a [f(x) + f(-x)] \, dx$

c. Hence, or otherwise, evaluate $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{1+\sin x} \, dx$

12. By using the substitution $x = 3 \sin \theta$, or

$$\text{otherwise, show that } \int \frac{x^2}{\sqrt{9-x^2}} \, dx = \frac{9}{2} \sin^{-1} \frac{x}{3} - \frac{x}{2} \sqrt{9-x^2} + k$$

13. Let s be a positive constant. Show that

$$\lim_{N \rightarrow \infty} \int_0^N t e^{-st} \, dt = \frac{1}{s^2}$$

14. Determine

a. $\int \cos^4 x \, dx$

b. $\int \sin^4 x \, dx$

c. $\int \sin^4 x \cos^4 x \, dx$

15. By using long division or otherwise, determine:

a. $\int \frac{x^2+x+1}{x+1} \, dx$

b. $\int \frac{x^3-2x^2+3}{x-2} \, dx$

c. $\int \frac{(x+1)^2}{1+x^2} \, dx$

16. Divide numerator and denominator by an appropriate factor to help determine

$$\int \frac{1}{x+\sqrt{x}} \, dx$$

17. a. Use a suitable substitution to help evaluate

$$\int_0^1 x(x-1)^5 \, dx$$

b. How could this integral have been evaluated using just algebraic manipulation?

18. Determine the primitive.

a. $\int \frac{1}{(1-x)\sqrt{x}} \, dx$

b. $\int \frac{x}{\sqrt{x+1}} \, dx$

19. Evaluate the integral.

a. $\int \frac{1}{(1+x^2)^{\frac{3}{2}}} \, dx$

b. $\int \frac{x^2}{\sqrt{4-x^2}} \, dx$

c. $\int \frac{1}{x^2\sqrt{25-x^2}} \, dx$

d. $\int \frac{1}{x^2\sqrt{1+x^2}} \, dx$

20. a. Determine $\int x \ln x \, dx$.

b. Hence find $\int x(\ln x)^2 \, dx$

21. Use trigonometric identities and then integration by parts to find:

a. $\int_0^{\frac{\pi}{2}} x \sin x \cos x \, dx$

b. $\int_0^{\frac{\pi}{2}} x \sin^2 x \, dx$

c. $\int_0^{\frac{\pi}{4}} x \tan^2 x \, dx$

d. $\int_0^{\pi} x^2 (\cos^2 x - \sin^2 x) \, dx$

22. a. Prove that $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1}{x(\pi-2x)} dx = \frac{2 \ln 2}{\pi}$

b. By using the substitution $t = a + b - x$,

prove that: $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$

c. Hence prove that: $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^2 x}{x(\pi-2x)} dx = \frac{\ln 2}{\pi}$

23. If $I_n = \int_0^{\frac{\pi}{2}} \sin^n 2x \cos^2 2x dx$ and $n \geq 1$

prove that $I_n = \frac{n-1}{n+2} I_{n-2}$ for $n \geq 2$. Hence use

this reduction formula to evaluate

$\int_0^{\frac{\pi}{2}} \sin^4 2x \cos^2 2x dx$

24. a. If $I_n = \int_0^t \frac{1}{(1+x^2)^n} dx$, $n = 1, 2, 3, \dots$, show

that $2n I_{n+1} = (2n-1) I_n + \frac{t}{(1+t^2)^n}$ for

$n = 1, 2, 3, \dots$

b. Hence find the value of I_3 in terms of t .

25. If $I_n = \int_1^e x (\ln x)^n dx$, $n = 0, 1, 2, 3, \dots$

a. Show that $I_n = \frac{e^2}{2} - \frac{n}{2} I_{n-1}$, $n =$

$1, 2, 3, \dots$

b. Evaluate $\int_1^e x (\ln x)^3 dx$

26. a. If $I_n = \int \frac{dx}{(x^2+1)^n}$ prove that $I_n =$

$\frac{1}{2(n-1)} \left[\frac{x}{(x^2+1)^{n-1}} + (2n-3) I_{n-1} \right]$

b. Hence evaluate $\int_0^1 \frac{dx}{(x^2+1)^2}$

27. a. If $I_n = \int x^3 (\log_e x)^n dx$, show that

$I_n = \frac{x^4}{4} (\log_e x)^n - \frac{n}{4} I_{n-1}$

b. Hence, or otherwise, evaluate

$\int_1^2 x^3 (\log_e x)^2 dx$

28. A curve is defined by the equation

$y = \frac{a}{2} \left(e^{\frac{x}{a}} + e^{-\frac{x}{a}} \right)$

a. Show that $1 + \left(\frac{dy}{dx} \right)^2 = \frac{y^2}{a^2}$

b. The arc length S between points $(0, a)$

and (x, y) of the curve is given by

$S = \int_0^x \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx$ (DO NOT

PROVE THIS).

Show that $S = \sqrt{y^2 - a^2}$.

29. Determine these logarithmic integrals.

a. $\int \frac{x}{1-x^2} dx$

b. $\int \frac{1+\sec^2 x}{x+\tan x} dx$

c. $\int \frac{\cos 3x}{1+\sin 3x} dx$

Answers

1. $\frac{1}{2} \ln 2$

2. $\sin 1$

3. $\frac{\pi}{8}$

4. $\ln(1 + \sqrt{2})$

5. $\ln(1 + \sqrt{2})$

6. $\frac{\pi}{4}$

7. $\frac{1}{2}$

8. $\ln 2$

9. $\frac{1}{2} \ln(x^2 + 1) - \tan^{-1} x$

10. $x^2 - x + \ln|x + 1|$

11. 2

14. a. $\frac{1}{32} (\sin 4x + 8 \sin 2x + 12x) +$

C ; b. $\frac{1}{32} (\sin 4x - 8 \sin 2x +$

$12x) + C$; c. $\frac{1}{1024} (24x -$

$8 \sin 4x + \sin 8x)$

15. a. $\frac{1}{2} x^2 + \ln(x + 1) + C$; b.

$\frac{1}{3} x^3 + 3 \ln(x - 2) + C$; c.

$x + \ln(1 + x^2) + C$

16. $2 \ln(1 + \sqrt{x}) + C$

17. a. $-\frac{1}{42}$; b. Begin by writing

$x = (x - 1) + 1$

18. a. $\ln \left(\frac{\sqrt{x}+1}{\sqrt{x}-1} \right) + C$; b. $\frac{2}{3} (x -$

$2) \sqrt{x+1} + C$

19. a. $\frac{x}{\sqrt{1+x^2}} + C$; b. $2 \sin^{-1} \frac{x}{2} -$

$\frac{1}{2} x \sqrt{4 - x^2} + C$; c. $-\frac{\sqrt{25-x^2}}{25x} + C$;

d. $-\frac{1}{x} \sqrt{1+x^2} + C$

20. a. $\frac{1}{4} x^2 (2 \ln x - 1) + C$; b.

$\frac{1}{4} x^2 (2 (\ln x)^2 - 2 \ln x + 1) + C$

21. a. $\frac{\pi}{8}$; b. $\frac{1}{16} (\pi^2 + 4)$; c.

$\frac{\pi}{4} - \frac{\pi^2}{32} - \frac{1}{2} \ln 2$; d. $\frac{\pi}{2}$

23. $\frac{\pi}{32}$

24. b. $I_3 = \frac{1}{8} \left(3 \tan^{-1} t + \frac{3t}{1+t^2} + \frac{2t}{(1+t^2)^2} \right)$

25. b. $\frac{1}{8} (e^2 + 3)$

26. b. $\frac{1}{4} \left(1 + \frac{\pi}{2} \right)$

27. b. $4 (\ln 2)^2 - 2 \ln 2 + \frac{15}{32}$

29. a. $-\frac{1}{2} \ln(1 - x^2) + C$; b.

$\ln(x + \tan x) + C$; c. $\frac{1}{3} \ln(1 + \sin 3x) + C$